



Savitribai Phule Pune University Fourth Year of Computer Engineering (2019 Course) 410251: Deep Learning		
Teaching Scheme:	Credit	Examination Scheme:
TH: 03 Hours/Week	03	In-Sem (Paper): 30 Marks End-Sem (Paper): 70 Marks
Prerequisite Courses: Machine Learning (410242)		
Companion Course: Laboratory Practice V(410254)		
Course Objectives: • To understand the basics of neural networks. • Comparing different deep learning models. • To understand the Recurrent and Recursive nets in Deep Learning • To understand the basics of deep reinforcement Learning models. • To analyze Types of Networks. • To Describe Reinforcement Learning.		
Course Outcomes: On completion of the course, student will be able to– CO1: Understand the basics of Deep Learning and apply the tools to implement deep learning applications CO2: Evaluate the performance of deep learning models (e.g., with respect to the bias-variance trade-off, overfitting and underfitting, estimation of test error). CO3: To apply the technique of Convolution (CNN) and Recurrent Neural Network (RNN) for implementing Deep Learning models CO4: To implement and apply deep generative models. CO5: Construct and apply on-policy reinforcement learning algorithms CO6: To Understand Reinforcement Learning Process		
Course Contents		
Unit I	Foundations of Deep learning	07 Hours
What is machine learning and deep learning?, Supervised and Unsupervised Learning, bias variance tradeoff, hyper parameters, under/over fitting regularization, Limitations of machine learning, History of deep learning, Advantage and challenges of deep learning. Learning representations from data, Understanding how deep learning works in three figures, Common Architectural Principles of Deep Network, Architecture Design, Applications of Deep learning, Introduction and use of popular industry tools such as TensorFlow, Keras, PyTorch, Caffe, Shogun.		
#Exemplar/Case Studies	Deep Mind, AlphaGo, Boston Dynamics	
*Mapping of Course Outcomes for Unit I	CO1	
Unit II	Deep Neural Networks(DNNs)	07 Hours

Introduction to Neural Networks :The Biological Neuron, The Perceptron, Multilayer Feed-Forward Networks , **Training Neural Networks** :Backpropagation and Forward propagation **Activation Functions** :Linear ,Sigmoid, Tanh, Hard Tanh, Softmax, Rectified Linear, **Loss Functions** :Loss Function Notation , Loss Functions for Regression , Loss Functions for Classification, Loss Functions for Reconstruction, **Hyperparameters** : Learning Rate, Regularization, Momentum, Sparsity, Deep Feedforward Networks – Example of XOR, Hidden Units, cost functions, error backpropagation, Gradient-Based Learning, Implementing Gradient Descent, vanishing and Exploding gradient descent, Sentiment Analysis, Deep Learning with Pytorch, Jupyter, colab.

#Exemplar/CaseStudies A Case Study for Music Genre Classification

***Mapping of Course** CO2

Outcomes for Unit II

Unit III Convolution Neural Network(CNN) 07 Hours

Introduction, CNN architecture overview, The Basic Structure of a Convolutional Network- Padding, Strides, Typical Settings, the ReLU layer, Pooling, Fully Connected Layers, The Interleaving between Layers, Local Response Normalization, Training a Convolutional Network

#Exemplar/Case Studies AlexNet, VGG

***Mapping of Course** CO3

Outcomes for Unit III

Unit IV Convolution Neural Network(CNN) 07 Hours

Recurrent and Recursive Nets: Unfolding Computational Graphs, Recurrent Neural Networks, Bidirectional RNNs, Encoder-Decoder Sequence-to-Sequence Architectures, Deep Recurrent Networks, Recursive Neural Networks, The Challenge of Long-Term Dependencies, Echo State Networks, Leaky Units and Other Strategies for Multiple Time Scales, The Long Short-Term Memory and Other Gated RNNs, Optimization for Long-Term Dependencies, Explicit Memory. **Practical Methodology**: Performance Metrics, Default Baseline Models, Determining Whether to Gather More Data, Selecting Hyper parameters.

#Exemplar/Case Studies Multi-Digit Number Recognition

***Mapping of Course** CO3

Outcomes for Unit IV

Unit V Deep Generative Models 08 Hours

Introduction to deep generative model, Boltzmann Machine, Deep Belief Networks, Generative adversarial network (GAN), discriminator network, generator network, types of GAN, Applications of GAN networks

#Exemplar/Case Studies GAN for detection of real or fake images

***Mapping of Course** CO4

Outcomes for Unit V

Unit VI Reinforcement Learning 07 Hours

Introduction of deep reinforcement learning, Markov Decision Process, basic framework of reinforcement learning, challenges of reinforcement learning, Dynamic programming algorithms for reinforcement learning, Q Learning and Deep Q-Networks, Deep Q recurrent networks, Simple reinforcement learning for Tic-Tac-Toe.

#Exemplar/Case Studies	Self driving cars, Deep learning for chatbots
*Mapping of Course Outcomes for Unit VI	CO5

Learning Resources

Text Books:

1. Goodfellow, I., Bengio, Y., Courville, A, “Deep Learning”, MIT Press, 2016.
2. Josh Patterson & Adam Gibson, “Deep Learning”
3. Charu Agarwal, “Neural Networks and deep learning”, A textbook
4. Nikhil Buduma, “Fundamentals of Deep Learning”, SPD
5. Francois chollet, “Deep Learning with Python”

Reference Books:

1. Richard S. Sutton and Andrew G. Barto, “Reinforcement Learning: An Introduction”
2. by Seth Weidman, “Deep Learning from Scratch: Building with Python from First Principles” O'Reilly
3. Francois Duval, “Deep Learning for Beginners, Practical Guide with Python and Tensorflow”

e-Books :

1. <http://csis.pace.edu/ctappert/cs855-18fall/DeepLearningPractitionersApproach.pdf>
2. https://www.dkriesel.com/_media/science/neuronale-netze-en-zeta2-1-col-dkriesel.com.pdf

MOOC Courses Links:

- <https://www.my-mooc.com/en/categorie/deep-learning>

CO\PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	-	-	3	-	-	-	-	-	-	2
CO2	3	2	2	2	1	-	-	-	-	-	-	1
CO3	3	2	2	2	2	-	1	-	-	-	-	1
CO4	1	2	1	1	2	-	1	-	-	-	-	1
CO5	2	2	3	2	2	-	-	-	-	-	-	1
CO6	1	2	2	2	2	-	-	-	-	-	2	-